



Knowledge Institute of Technology

Centre for Career Development & Training (CDT)

Summative Assessment I (Pen & Paper Mode)

Smartcliff & Silicon (SM – 1,2&3 and SI – 1,2&3)

Date: 29.06.2026 Duration: 1 Hr. 15 Mins.

Max.Marks: 100 Batch : 2027 / IV Year

Section – A (60 Minutes)

Programming Questions (80 Marks)

1. **The royal baker needs to prepare a special cake. Each ingredient has a specific weight. The baker wants to know the total weight of the ingredients and the heaviest ingredient used. (10 Marks)**

Sample Input

4

1.2 0.5 2.3 1.8

Sample Output

5.8 2.3

2. **A streaming service maintains a collection of movie titles without duplicates. (20 Marks)**

Create a class MovieCollection and implement the following methods.

```
void addMovie(String movie)
void removeMovie(String movie)
boolean containsMovie(String movie)
int movieCount()
Set<String> getMovies()
```

Sample Input

```
addMovie("Avatar");
addMovie("Inception");
addMovie("Avatar");
```

```
print(movieCount());
print(getMovies());
print(containsMovie("Titanic"));
```

Sample Output

2

[Avatar, Inception]

false

Test Case 2 :

Sample Input

```
addMovie("Avatar");
addMovie("Titanic");
addMovie("Joker");
```

```
removeMovie("Titanic");
```

```
print(containsMovie("Titanic"));
print(movieCount());
print(getMovies());
```

Sample Output

false

2

[Avatar, Joker]

3. **Given a string consisting only of bracket characters (,), { , }, [,], determine whether the brackets are correctly matched — every opening bracket has a corresponding closing bracket of the same type, and brackets are closed in the correct order (properly nested). (30 Marks)**

Input Format:

- A single line containing the string S (consisting only of bracket characters).

Output Format:

- Print True if the string is balanced, otherwise print False.

Example 1:

Input: ({})

Output: true

Example 2:

Input: ([)]

Output: false

(Explanation: [is closed by) before its matching] appears — incorrect nesting.)

Constraints:

- $1 \leq |S| \leq 10^4$
- S contains only the characters (,), {, }, [,]

**4. Given a singly linked list, perform the following operations on it in order:
(40 Marks)**

- Insert a new node with a given value immediately after a specified position.
- Delete the node at a specified position.
- Reverse the entire linked list.

Return the final linked list after all three operations have been applied.

Input & explanation:

3 → number of initial nodes

1 2 4 → the 3 initial values

3 2 → insert value 3 after position 2

2 → delete the node at position 2 (from the updated 4-node list)

Output:

4 3 1 → final list after insert, delete, and reverse

(Contd. to Section B)

Section – B (15 Minutes)
Aptitude Questions (10 * 2 = 20)

- 1) A car travels from City P to City Q at a speed of 60 km/hr and returns from Q to P at a speed of 40 km/hr. What is the average speed of the car for the entire journey?
A) 45 km/hr B) 48 km/hr C) 50 km/hr D) 52 km/hr E) 55 km/hr
- 2) A train 240 metres long is running at a speed of 72 km/hr. In how many seconds will it cross a platform 360 metres long?
A) 20 seconds B) 24 seconds C) 28 seconds D) 30 seconds E) 36 seconds
- 3) A boat covers 32 km downstream in 4 hours and the same distance upstream in 8 hours. What is the speed of the boat in still water?
A) 3 km/hr B) 4 km/hr C) 5 km/hr D) 6 km/hr E) 8 km/hr
- 4) A sum of money amounts to ₹9,440 in 3 years and ₹10,400 in 5 years at simple interest. What is the principal sum?
A) ₹7,800 B) ₹7,900 C) ₹8,000 D) ₹8,060 E) ₹8,180
- 5) Find the compound interest on ₹15,000 for 2 years at 10% per annum, compounded annually.
A) ₹3,000 B) ₹3,100 C) ₹3,150 D) ₹3,200 E) ₹3,300
- 6) In an examination, a student scored 35% marks and failed by 12 marks. Another student scored 50% marks and obtained 18 marks more than the minimum passing marks. What is the maximum mark of the examination?
A) 180 B) 190 C) 195 D) 200 E) 210
- 7) A shopkeeper marks an article 40% above its cost price and then allows a discount of 15% on the marked price. What is his overall profit percentage?
A) 16% B) 18% C) 19% D) 21% E) 24%
- 8) Five friends — Manoj, Neha, Om, Priya, and Rahul — are ranked based on their salaries. Neha earns more than Om but less than Priya. Rahul earns more than Priya. Manoj earns less than Om. Who earns the highest salary?
A) Manoj B) Neha C) Om D) Priya E) Rahul
- 9) Six people — J, K, L, M, N, and O — sit in a row facing north. K sits third from the left end. M sits immediately to the right of K. L sits at one of the extreme ends. N sits second from the right end. J sits immediately to the left of O. Who sits at the left extreme end?
A) J B) K C) L D) N E) O
- 10) In a class of 40 students, Aman ranks 14th from the top. Sonia ranks lower than Aman, with exactly 5 students positioned between them. How many students rank below Sonia in the class?
A) 18 B) 19 C) 20 D) 21 E) 22
